

**Homework 9**

1. Use any test to determine whether the series is absolutely convergent, conditionally convergent, or divergent:

(a)  $\sum_{n=1}^{\infty} \frac{(-1)^n}{n \ln(n+1)}.$

(b)  $\sum_{n=1}^{\infty} \left( \frac{\ln(n)}{n} \right)^3.$

(c)  $\sum_{n=1}^{\infty} \frac{e^{2n}}{6^{n-1}}.$

(d)  $\sum_{n=1}^{\infty} \ln \left( \frac{n}{n+1} \right).$

2. Consider the following series:

$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{n^3 3^n}$$

- (a) Find the series' radius of convergence.  
(b) Find all the values  $x \in \mathbb{R}$  such that the series converges.

3. For the following function, find the Maclaurin series:

$$f(x) = \frac{1}{1+x}.$$

4. Use the known series and substitution to find the Taylor series at  $x = 0$  of the following function

$$f(x) = \frac{1}{1 + \frac{3}{4}x^3}.$$

5. Use power series operations to find the Taylor series at  $x = 0$  for the function

$$f(x) = \frac{x^2}{1-2x}.$$

6. Use the Taylor series at  $x = 0$  of  $f(x) = e^x$  to find an approximation of  $\sqrt{e}$  with an accuracy of  $10^{-5}$ .  
7. How close is the approximation  $\sin x \approx x$  when  $|x| < 10^{-3}$ ? For which of these values of  $x$  is  $x < \sin x$ ?  
8. The approximation  $e^x \approx 1 + x + (\frac{x^2}{2})$  is used when  $x$  is small. Use the Remainder Estimation Theorem to estimate the error when  $|x| < 1$ .